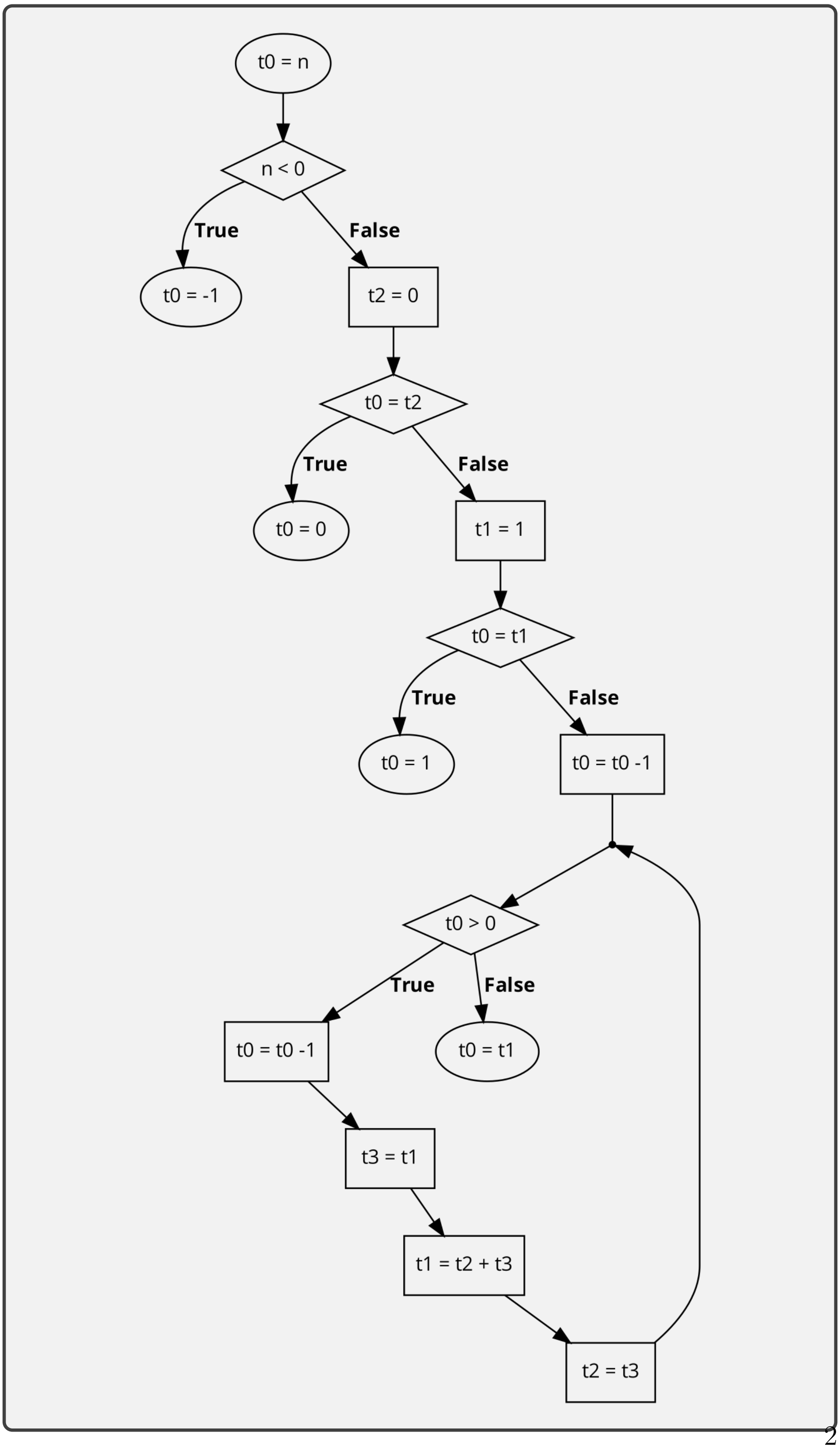
Exercise Sheet

Task 1 – Fibonacci Sequence

The Fibonacci sequence is a numerical series where each number in the sequence is decided by adding the two previous numbers from it.
Let $F(n)$ be the nth Fibonacci number, then $F(n) = F(n-2) + F(n-1)$
Additionally $F(0) = 0$ and $F(1) = 1$

Implement a program that computes the nth Fibonacci number, where n is an integer in register t0. The return value should also be put into register t0. If the number in register t0 is negative, your program should return -1. Make sure to include an overflow check and comments.

a) Draw a Flow Chart to plan your implementation


t0 = n
n < 0
True
False
t0 = -1
t2 = 0
t0 = t2
True
False
t0 = 0
t1 = 1
t0 = t1
True
False
t0 = 1
t0 = t0 -1
t0 > 0
True
False
t0 = t0 -1
t0 = t1
t3 = t1
t1 = t2 + t3
t2 = t3

b) Implement the Algorithm

```
# Load n and check if n > 0
li t0, 7
bgt zero, t0, nNegative

# Check for cases n=0 and n=1
li t2, 0
beq t0, t2, nZero
li t1, 1
beq t0, t1, nOne
addi t0, t0, -1

# Compute F(x) = F(x-2) + F(x-1)
loop:
    beq t0, zero, copyFn
    andi t3, t1, -1
    add t1, t2, t3
    andi t2, t3, -1
j loop

#Return value for n < 0
nNegative:
    li t0, -1
    j end

#Return value for n=0
nZero:
    li t0, 0
    j end

#Return value for n=1
nOne:
    li t0, 1
    j end

#Return value for n > 1
copyFn:
    andi t0, t1, -1

#Jump mark for end of program
end:
    nop
```